JMAN GROUP

CRM Churn Prediction Pipeline for pre-renewal period

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THE SUNDAY TIMES Hundred 2024 BRITAIN'S FASTEST-GROWING PRIVATE COMPANIES

Project Context | Problem Statement & Objective

Objective: Predict contractor churn before renewal so the team can intervene proactively and reduce revenue loss

The Business Problem

Contractor churn is costly and largely reactive. Renewals teams intervene only after a contractor shows disengagement signals — or worse, after they have already lapsed.

Without a predictive signal, the business cannot:

- Prioritize which contractors need outreach
- Allocate renewal team capacity effectively
- Identify the root cause of churn in advance

Objective

Predict whether a contractor will churn (not renew) or be retained — using pre-renewal data only. No post-outcome information used.

Target Variable

`churn_label`

1 → Churned (did not renew)

0 → Won (renewed)

Binary Classification

Primary Metric

PR-AUC & F1

Precision-Recall AUC & F1

Robust to class imbalance. Focuses on the minority churn class.

Pragatheesh-I/06_Pre-renewal_churn

Use Case | Data Sources & Key Signals

Use Case: Score each contractor BEFORE their renewal date so the renewals team can proactively intervene — reducing churn through targeted, timely outreach.

Billings

KEY SIGNAL

Core renewal features + churn label

One row per renewal cycle — prices, bands, tenure, audit and membership status. Unique key: Co_Ref + Renewal_Date

Renewal Calls

KEY SIGNAL

Churn intent, complaints, price discussions

Call centre recordings filtered to 14-day pre-renewal window. Includes desire-to-cancel, competitor mentions, pricing signals

CC Calls

KEY SIGNAL

Sentiment, platform issues, financial hardship

General contact centre calls in the 14-day pre-renewal window. Contractor sentiment scoring, flag columns for 13 signal types

Emails / CRM

KEY SIGNAL

Accreditation progress, payment intention

CRM email notes from pre-renewal period. 19 binary/ternary flag columns covering satisfaction, chase count, membership level

All interaction data filtered to the 14-day pre-renewal window to prevent post-outcome leakage

ML Pipeline | 5-Stage Architecture

1 Data Preprocessing	2 Merge	3 EDA & Feature Eng.	4 Feature Selection	5 Model Building
INPUT	INPUT	INPUT	INPUT	INPUT
Raw CSV files	4 preprocessed CSVs	final_merged.csv	df_model.csv	final_churn_ dataset.csv
OUTPUT	OUTPUT	OUTPUT	OUTPUT	OUTPUT
4 cleaned files	final_merged.csv	df_model.csv	final_churn_ dataset.csv	Trained model + eval charts
NOTEBOOK	NOTEBOOK	NOTEBOOK	NOTEBOOK	NOTEBOOK
4 cleaning notebooks	merge.ipynb	analysis.ipynb	FeatureSelection .ipynb	hptuning_ mbuilding.ipynb

Data Preprocessing | Steps Taken & Outcomes

Objective: Clean each of the 4 raw data sources independently using domain knowledge — not generic imputation — before merging

Steps Taken

1. **Drop irrelevant columns**
   Removed 100% null, zero-variance and duplicate cols (e.g. Last_Years_Date_Paid, DateTime_Out, Billings_Contractor)
2. **Fix data types & errors**
   Date strings → datetime64; boolean conversions; Renewal_Year=2050 corrected; Tenure_Years recomputed from Registration_Date
3. **Create churn label**
   churn_label = 1 (Churned), 0 (Won). Open rows & 6 noise statuses excluded before label creation
4. **Domain-driven null fills**
   0 for connections (94% null = no connections); current flag for Proforma nulls; 'New_Customer' for Last_Band
5. **Group & encode categoricals**
   31 Prospect_Status values → 10 groups; 31 Audit_Status → 7 groups; CC/Email flags standardised to Yes/No/Not Mentioned
6. **Filter interaction data**
   CC Calls & Renewal Calls filtered to 14-day pre-renewal window only (0 ≤ days_to_renewal ≤ 14)

Outcomes at Each Stage

✓	4 source-specific, cleaned datasets ready for merging (dataset/preprocessed/)
✓	Churn label created with 6 noise statuses and all 'Open' rows excluded
✓	All date columns correctly typed; data entry errors fixed at row level
✓	Domain-meaningful null fills applied — not generic mean/median imputation
✓	Categoricals grouped into interpretable, compact, domain-validated sets
✓	Interaction data (calls/emails) window-filtered for leakage prevention
✓	Tri-state encoding preserves 'no call happened' as distinct from 'signal absent'

Merge & EDA | Joining Data & Discovering Patterns

Stage 2 — Merge

1. **Create join key**

   year_key = Co_Ref + '_' + year created across all 4 tables as the unified join key

2. **Aggregate interactions**

   Binary flags → max (1 if ANY interaction had signal); counts → count; scores → mean, min, last_value; categoricals → meaningful_mode

3. **Left join on Billings**

   All interaction tables joined to Billings. Post-merge: count cols → 0; interaction cols → 0; categorical cols → 'No_Interaction'

Output: final_merged.csv — single joined dataset ready for EDA

Stage 3 — EDA Steps

- Leakage detection — Closed_Date, Prospect_Status, Total_Net_Paid dropped
- Outlier handling — IQR; row-level removal for data entry errors only
- NZV lift analysis — columns with lift > 2× retained despite low frequency
- Correlation removal; price_change & price_increase_pct engineered

EDA Key Findings

Feature	Finding
Band	Band A: ~22% churn vs Band J: ~5% — inverse relationship
Tenure Group	Year 1: ~21% churn → Year 4+: ~7% — longer tenure = lower churn
Membership	Member Only: ~29% churn vs Accredited: ~6%
Audit Status	Initial_Failed: ~34% churn vs Accredited: ~6%
Payment Method	BACS: ~0% churn; UNKNOWN: ~100% — strongest signal
price_change	Churned customers experienced larger price increases
rc_total_calls	More renewal calls correlated with higher churn (escalation proxy)
Auto Renewal	Auto-renewal flag strongly associated with retention

Feature Selection | 3-Step Process — Tree Rankings + Business Override

1 Tree-Based Importance

OUTCOME
Rankings from both models produced

- Random Forest — Mean decrease in impurity (Gini). n_estimators=300, max_depth=8, class_weight=balanced
- XGBoost — Total gain across splits. n_estimators=300, max_depth=6, scale_pos_weight=imbalance ratio
- 80/20 stratified train/test split used for importance evaluation

2 Combined Rank Aggregation

OUTCOME
52 tree-selected features

- avg_rank = (rf_rank + xgb_rank) / 2 — averaged to reduce single-model bias
- Top 52 features by average rank form the tree-selected set
- Avoids selecting features important to only one model's inductive bias

3 Business-Critical Feature Override

OUTCOME
Final set: tree (52) + business overrides

- 31 domain-validated features force-added regardless of tree importance rankings
- Covers: Tenure, Band, Pricing (price_change, price_increase_pct, Gross, Amount)
- Renewal Scores, Connection data, Call signals (desire to cancel, churn category)

Final dataset saved with identifier columns (year_key, Co_Ref, Renewal_Year) + features + churn_label

Model Building | Baseline → HP Tuning → Ensemble

1 Baseline Models

- Logistic Regression (+ StandardScaler)
- Random Forest — class_weight=balanced
- XGBoost — scale_pos_weight
- LightGBM — class_weight=balanced
- CatBoost — auto_class_weights=Balanced

2 Hyperparameter Tuning (Optuna)

- Bayesian optimisation — 20 trials/model
- 5-fold Stratified CV on training set
- Maximise mean PR-AUC across folds
- XGBoost: n_est 200-600, depth 4-10, lr 0.01-0.3
- LightGBM: num_leaves 20-150, feature_frac 0.6-1.0

3 Ensemble Methods

- Soft Voting — probability averaging across models
- Stacking — Logistic Regression meta-learner
- Base predictions via 5-fold CV (prevents meta-leakage)
- Best model across ALL stages selected as final
- Models ranked by test set PR-AUC

Threshold Optimisation — F2-Score

Rather than using 0.5 threshold, optimal threshold found by maximising F2-Score on a held-out validation fold. Thresholds 0.10→0.90 evaluated.

F2 = (5 × precision × recall) / (4 × precision + recall)

Recall weighted 2× — missing a churner costs more than a false alarm

Time-Based Train / Test Split

SET	CRITERION	PURPOSE
Train	Renewal_Year < 2025	Historical data — model learns from past
Test	Renewal_Year ≥ 2025	2025 renewals — unseen future data

Simulates real-world deployment — train on past, predict future. Prevents data leakage.

Assumptions | Key Design Decisions & Rationale

Design Decision	Rationale
PR-AUC as primary metric	Class imbalance makes accuracy and ROC-AUC misleading. PR-AUC focuses on the minority churn class where business value lies
F1-Score for threshold tuning	Missing a churner (false negative) and raising a false alarm (false positive) are treated as equally important — precision and recall are balanced.
Time-based train/test split	Prevents leakage and simulates deployment: model is trained on past data and evaluated on future (2025) renewals
Leave outliers untreated	Tree-based models are robust to outliers. Transformations would remove meaningful business signals (e.g. high billing amounts)
Lift-based NZV retention	Rare events with lift > 2× carry genuine predictive power (e.g. rc_Explicit_Switching_Intent) despite low frequency
Tri-state interaction encoding	Distinguishes 'no call happened' from 'call happened but signal absent' — critical distinction for interaction features
Business-critical feature override	Ensures 31 interpretable, domain-validated features enter the model even if not top-ranked by automated tree importance
Domain-specific null fills	E.g., 0 for missing connections (no connections present) vs current flag for proforma flags (reliable proxy)

Solution Architecture | End-to-End Pipeline Design

Raw Data (4 Sources)	Preprocessing & Merge	EDA & Feature Eng.	Feature Selection	Model Training	Scored Contractors
Billings · Calls · Emails · CRM	Clean · Join · Encode	Leakage · Outliers · Lift	Tree Rank + Override	3-Stage + Ensemble	Risk Score per Renewal

Leakage Prevention

- Time-based split (train <2025, test ≥2025)
- Pre-renewal window enforcement (14 days)
- 3 systematic leakage checks before training
- Closed_Date, Prospect_Status, payment cols dropped

Class Imbalance Handling

- PR-AUC & F1 as primary metric (not accuracy)
- class_weight=balanced / scale_pos_weight used
- F1-Score threshold optimisation at inference
- Minority class (churn) focus throughout

Ensemble Strategy

- Stacking — LR meta-learner on 5-fold CV base
- Prevents meta-level leakage via CV in stacking
- Best model selected across all 3 stages

Interpretability

- 31 business-critical features force-included
- Domain-driven null fills (not generic imputation)
- Tri-state encoding for interaction features
- Lift-based NZV retention for rare but signal-rich cols

Final Results | Evaluation Metrics, Model Selection & Business Outputs

PR-AUC	ROC-AUC	Recall	Precision	F1-Score
Primary Metric (Minority Class Focus)	Secondary (Overall Discrimination)	Churn Capture Rate (Sensitivity)	Flag Accuracy (of Flagged = True Churn)	**Balanced (Equal Weight to Precision & Recall)**

Model Selection Process

Stage 1: 5 Baseline Models Ranked by PR-AUC → **Top 2 → Optuna Tuning 20 trials, 5-fold CV** → **Ensemble Methods (Soft Voting + Stacking)** → **Best Model Selected across all stages**

Business Outputs

- Churn risk score for each contractor before renewal date
- Confusion matrix — TP/FN/FP/TN with capture rate & flag accuracy
- Precision-Recall curve with operating point at optimal threshold
- Business-readable summary for renewals team intervention planning

Key Terminology

Lift:	Churn rate in segment ÷ overall churn rate
Tri-state:	No_Interaction / False / True → two binary cols
NZV:	Near-zero variance — one value in ≥99% rows
Stacking:	Meta-learner trained on base model predictions

Dashboards and Report

1. Summary Dashboard
2. Customer Risk List
3. Churn Drivers

Summary Dashboard

Pre Renewal Churn Analysis

Churn
All

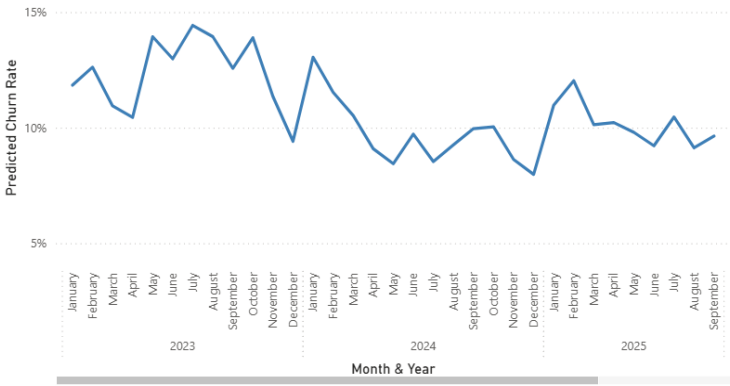
Band
All

Renewal Year
All

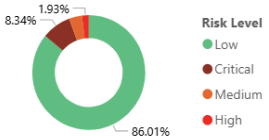
Renewal Date
1/1/2023
5/1/2026

112K	10M	12K	89.60%
Total Customers	Revenue At Risk	Predicted_Churn_Count	Retention Rate

Predicted Churn Rate by Year and Month



Total Customers by Risk Level



Risk Level by Prospect Count

Risk Level	P1	P2	P3	P4
Critical	2654	6695		
High	884	1283		
Medium		1953	2213	
Low				96399

Customer Risk List

Pre Renewal Churn Analysis

Risk Level
All

Propect Category
All

Band
All

Tenure Group
All

Connection Gro...
All

3,538 | £4,754,876
P1 - Highest Priority Accounts

9,931 | £8,167,324
P2 - High Priority Accounts

2,213 | £1,496,663
P3 - Medium Priority Accounts

96,399 | £107,267,053
P4 - Low Priority Accounts

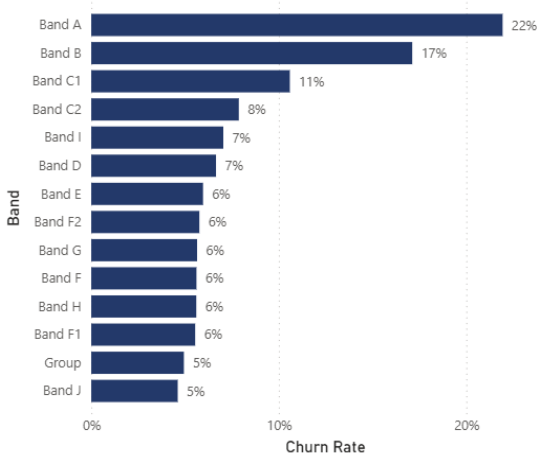
Customer ID	Risk Level	Prospect Category	Churn Probability	Amount	Band	Tenure Years	Connection Group	Renewal Month & Year
AA0106	Critical	P1	98.00%	£1,119.0	Band D	15	3	February, 2024
AB2021	High	P1	52.23%	£1,809.0	Band F2	7	Independent	June, 2024
AB2865	Critical	P1	77.22%	£1,019.0	Band C2	2	1	August, 2024
AB6762	Critical	P1	100.00%	£1,099.0	Band C2	1	Independent	April, 2025
AB8848	High	P1	52.18%	£1,849.0	Band F1	1	1	July, 2025
AB9976	Critical	P1	99.98%	£1,309.0	Band D	1	Independent	July, 2025
AC1671	Critical	P1	94.69%	£1,849.0	Band F1	4	Independent	July, 2025
AC2155	Critical	P1	100.00%	£1,069.0	Band C2	2	Independent	April, 2025
AC6856	Critical	P1	99.97%	£2,389.0	Band F1	2	Independent	November, 2025
AC9030	Critical	P1	99.23%	£1,149.0	Band D	3	Independent	January, 2024
AD0310	Critical	P1	99.04%	£1,859.0	Band F2	1	2	July, 2024

Churn Drivers

Pre Renewal Churn Analysis

10.71%	29.65%	9.12%
Churn Rate	New Customer Churn Rate	Existing Customer Churn Rate

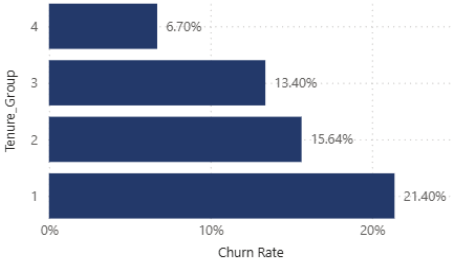
Churn Rate by Band



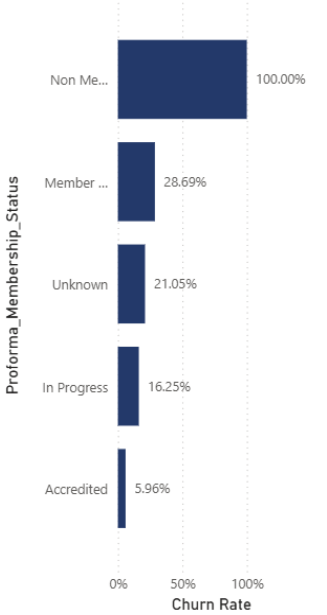
Filter by Band

All

Churn Rate by Tenure_Group



Churn Rate by Proforma_Membership_Status



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